

Christal Chen

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EDUCATION

Princeton University

Princeton, NJ

Bachelor of Science in Engineering, Electrical and Computer Engineering

Aug. 2024 – May 2028

- **GPA:** 3.77/4.00
- **Intended minor:** Computer Science
- **Relevant coursework:** Contemporary Logic Design, Electronic Circuit Design, Introduction to Machine Learning, Information Signals, Algorithms and Data Structures, Introduction to Programming Systems, Linear Algebra

EXPERIENCE

Digital Design Intern

June 2026 – Present

SiFirst Technologies

Yongin, South Korea

- Designed and implemented a synthesizable full-duplex 4-wire SPI interface in Verilog, supporting all four SPI modes and configurable data transfers.
- Focused on clean RTL code and systematic datapath and controller design to enhance scalability, extending SPI functionality from single-byte to multi-byte transfers.
- Debugged RTL through waveform analysis and simulation-based verification, ensuring correct serial transmit/receive behavior across multiple SPI modes and read/write operations.
- Documented design through block diagrams and technical reports, and communicated design decisions within an international engineering team environment.

PROJECTS

PUncC: 16-bit RISC Microprocessor Design | Verilog, Vivado, FPGA

Nov 2025 – Dec 2025

- Designed and implemented a 16-bit RISC microprocessor in Verilog as a coursework project, using a custom LC3 instruction set architecture supporting arithmetic, logic, and control flow operations.
- Engineered a datapath and FSM controller to implement a three-stage fetch-decode-execute cycle, integrating a 16-bit ALU, register file, and synchronous memory.
- Verified RTL functionality through Vivado simulation and modular testbenches, validating instruction-level correctness and executing programs such as GCD.

RESEARCH

Undergraduate Researcher

Nov 2025 – May 2026

Princeton Safe Robotics Lab

Princeton, NJ

- Enhanced a gameplay-based safety filter for autonomous robotic systems by implementing branching rollouts for multi-scenario evaluation.
- Trained an agent with adversarial reinforcement learning and ran safety-validation experiments in MuJoCo with a simulated Unitree Go2 quadruped, experimentally determining optimal hyperparameter values.
- Utilized Princeton's Della HPC cluster for model training and Git for version control while performing statistical analysis to quantify safety filter performance.

ACTIVITIES

Embedded Software Engineer

Sep 2025 – May 2026

Princeton Electric Speedboating

Princeton, NJ

- Developed an ESP32-based CAN bus interface to communicate with a battery management system for an electric speedboat.
- Implemented CAN message polling, request/response handling, and data caching for reliable real-time telemetry.
- Parsed CAN payloads and develop message decoders for reliable extraction of battery and motor telemetry, including state of charge, temperatures, motor speed, and current.

TECHNICAL SKILLS

Languages: Verilog, C/C++, Python, TCL, Java, R, SQL

Hardware: RTL Design, Digital Logic, FSM Design, Datapath Design, ESP32, CAN Bus

Tools: Vivado, Git, Vim, MuJoCo